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Fabrication of titania/hydroxyapatite composite granules for photo-catalyst

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ABSTRACT

The titania/hydroxyapatite composite granular photo-catalyst with novel microstructure was fabricated by the process based on the liquid immiscibility effect and followed by precalcination and hydrothermal treatment from commercially available powders of α-Tri-calcium phosphate and TiO_2. XRD, SEM, BET, optical microscopy and UV–vis spectrophotometer were applied to characterize the prepared photo-catalyst. Microstructure analysis indicated that the granule was weaved by rod-shaped hydroxyapatite crystals whose surface was covered by nano-sized TiO_2. In the composite granules, the active surface of anatase was retained effectively. With the hybridization of TiO_2 and HAp, a 16-nm blue-shift of absorption edge could be observed and the crystallinity of anatase could be enhanced by precalcination. The granules with the rod-shaped hydroxyapatite crystals performing as scaffold work as three-dimensional high porous, size-controllable small reactor. The phase and microstructure transformation of the granule before and after hydrothermal treatment was investigated and its decomposition ability was evaluated by using Methylene blue as a target pollutant compound.

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1. Introduction

One of the most important challenges for scientists is to develop efficient methods to control environmental pollution. The TiO_2 photo-catalyst shows very high activity for photo-oxidation of organic compounds either in air or in liquid solution. And it has been proved to be a kind of panacea for the removing of organic compounds [1]. It is close to be an ideal photo-catalyst in several aspects due to many prominent properties, such as high chemical stability, high light damaging resistance, commercially available and relatively inexpensive. The photo-generated holes are highly oxidative and photo-generated electrons are reductive enough to produce superoxide from dioxygen [2]. For practical application, however, there are some disadvantages remained to be resolved [3–5]. For instances, it is difficult to filter or recycle fine TiO_2 powder after the industrial process of purifying contaminated water; it is also hard to effectively immobilize fine catalytic particle and difficult to efficiently utilize the sunlight.

In order to avoid the use of fine TiO_2 powder directly, because of the involvement of later separation from water, numerous researchers worked on ways to immobilize TiO_2 particle. For example, Matthews [6] firstly reported on the preparation of TiO_2 film via sol–gel method. This idea and other methods also have been practiced by others [7–9]. So the design and development of highly efficient photo-catalytic materials is the key point of photo-catalysis and has attracted a great deal of attention for its potential in addressing environmental concerns.

Hydroxyapatite, $Ca_{10-x}(HPO_4)_x(PO_4)_{6-x}(OH)_{2-x}$ (x varies from 0 to 2) (HAp) is known as not only a primary constituent of the biological hard tissues but also adsorbent and catalysts [10]. It has been used for chromatographic columns by taking advantages of its ability to selectively adsorb protein and for controlled release carries of protein such as bFGF [11]. Calcium phosphate system and its compounds have been studied for a long time from different aspects. In this system, HAp was proved to be the most stable phase in most cases [12]. The formation, properties and the applications of HAp also have been reported in many works. The authors [13,14] reported various kinds of HAp materials prepared by the unique hydrothermal method. By this method, rod-shaped hydroxyapatite was successfully fabricated and its properties were researched widely. Hydrothermal method has also been proved to be an environmentally friendly way to control the morphology of HAp crystals by others [15].

Nonami et al. [16] reported their work of apatite-coated titanium dioxide photo-catalyst for air purification. In this paper, the apatite film covered parts of the surface of fine dense TiO_2 powders by biomimetic processing. The composite catalyst showed high adsorption ability and partly solved the compatibility problem with the polymer substrate, but did not enhance the decomposition ability to acetaldehyde distinctively. Yamashita

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et al. [17] researched the liquid-phase photo-catalytic oxidative degradation of 1-octanol in oxygenated acetonitrile on highly dispersed titanium oxide catalysts anchored on the surface of transparent vycor glass and compared it with the obtained standard TiO_2 powder. Nakajima et al. [18] prepared porous TiO_2 film by controlling phase separation conditions of Ti-alkoxide on the surface of HAp film which was prepared by spray pyrolysis deposition process. Yamashita et al. [19] fabricated TiO_2 photo-catalyst by an ionized cluster beam method on porous vycor silica glass and activated carbon fibers. All these results indicated that high active surface of anchored TiO_2 play an important role in the enhancement of its decomposition ability. But most of these works used complicate equipments and processes or various organic compounds as starting materials some of which are very harmful to human body and produce environmental problems as a side-effect.

The efficiency of photo-catalytic decomposition could be enhanced by anchoring photo-active catalysts on suitable support particularly that with large surface area [19]. The design of titanium oxide photo-catalysts anchored or embedded onto support materials with large surface area, which could condense dilute polluted substances, would be of great significance. It is not only to avoid the filtration and suspension of small photo-catalyst particles, but to obtain higher efficiency. In this paper, our work was concentrated on the design and actualization of titania/hydroxyapatite composite photo-catalyst. The evolution of its microstructure was analyzed in details. And its photo-catalytic ability was evaluated by the decomposition of Methylene blue as a target pollutant compound.

2. Experimental

2.1. Materials

Commercially available powders of α-Tri-calcium phosphate (α-$Ca_3(PO_4)_2$ (α-TCP): Taihei Chemical Industrial Co., Ltd., Japan) and TiO_2 (Item No. ST-01, Ishihara Sangyo, diameter 7 nm) were used as starting materials for the preparation of hydroxyapatite or titania/hydroxyapatite composite. To fabricate the granules, the gelatin (Wako Pure Chemical Industries, Ltd., Japan) and vegetable oil were used. In this paper, Methylene blue (Extra pure reagent, Merck KGaA, Darmstadt, Germany) was applied to evaluate the decomposition ability of fabricated photo-catalyst.

2.2. Procedure

The method to prepare porous spherical granules is based on liquid immiscibility effect [20]. Aqueous gelatin solution with the concentration of 10 wt.% was prepared by dissolving gelatin in distilled water at the temperature ranging from 45 °C to 65 °C. Titania (ST-01) and α-TCP were mixed thoroughly in mortar in the proportion of 30–70 wt.%. The mixed fine powder was added to the above solution in the proportion of 1 g powder per 2.5 ml gelatin solution. The suspension was ultrasonically dispersed and the temperature was kept by heating the water in the tank of the ultrasonic equipment. After that, the slurry was dropped and dispersed in vegetable oil in a beaker by stirring the paddle wheel at 300 rpm at 40 °C for 10 min. The size of granules can be conveniently controlled by changing stirring speed and oil temperature as reported in literatures [20–22]. The stirring results in granule formation due to the surface tension force and the immiscibility of oil and gelatin solution. The precipitated beads were washed in ethanol, filtered and dried in air at the ambient temperature of 4 °C. The same procedure could be repeated to produce HAp granules with the starting material of α-TCP only.

These gelatin-bonded granules were precalcined 1 h initially at 350 °C, 450 °C, 550 °C, 650 °C and 850 °C, respectively for titania/α-TCP composite and at 1200 °C for granules containing α-TCP only for burning off the gelatin. Lower temperatures were selected to remove the binder in the composite for the purpose of keeping the anatase phase unchanged. For comparison, the ST-01 nano-powder was calcined at 550 °C for 1 h (denoted as ST-01-550). After the removal of binder, 0.5 g granule was put in a 105-cm^3 autoclave with 30 cm^3 pure water, and then it was hydrothermally treated at the temperature of 120 °C under the saturated vapor pressure for 24 h. The mixed fine powders of α-TCP and ST-01 were pressed under the pressure of 200 MPa in a die with the diameter by 8 mm. The compressed photo-catalytic powder was gained by crushing the hydro-thermally treated pellets under the same condition for the fabrication of granules.

2.3. Characterization

The shape of the fabricated granules was investigated with optical microscope VHX-100 (Keyence Co., Ltd., Japan) and scanning electronic microscopy (SEM) (JEOL, JSM-T300, Japan). The X-ray diffraction (XRD) patterns obtained on an X-ray diffractometer (Rigaku RINT 2200VL, Japan, operating at 40 KV, 40 mA, CuK_α radiation, λ = 0.15406 nm) were used to determine the present phase and its crystallinity qualitatively. The morphologies of granular samples before and after hydrothermal treatment were observed using SEM. The Brunauer–Emmett–Teller surface area (S_{BET}) of the samples was analyzed by nitrogen adsorption in a Micromeritics ASAP2020 nitrogen adsorption apparatus (USA). All samples were degassed at 180 °C prior to nitrogen adsorption measurements. The BET surface area was determined by multi-point BET method using the adsorption data in the relative pressure (P/P_0) range from 0.05 to 0.35. The diffused reflectance spectrum (DRs) was recorded by UV–vis spectrophotometer (V-570, Jasco, Japan) with the attachment of integrating sphere and powder holder.

2.4. Measurement of photo-catalytic activity

Methylene blue, a widely used dye in textile printing industry, was used as the target compound in this study to evaluate the photo-catalytic activity of the titania/HAp composite. The primary concentration of the model contaminate chemical, Methylene blue, was 5 ppm for each experiment. 0.15 g titania/HAp composite catalyst or the equivalent amount of pure TiO_2 (ST-01 or ST-01-550) was suspended in 20 ml Methylene blue solution under continual stirring at 300 rpm. The vessel was enveloped in a thick container, so other lights except the experimental ultraviolet source irradiated from a handy UV lamp LUV-6 (As One Co., Ltd., 9 W) was blocked. The dominating wavelength of the irradiation was 365 nm after the cutting off by a filter with the size of 154 mm × 47 mm. The ultraviolet source was put on the top of the vessel with the distance of 10 cm. Before the ultraviolet irradiation was applied, the suspension was initialized with the adsorption of Methylene blue by catalyst. The period of initialization was 1 h for each sample with continuous stirring. In most cases, the equilibrium was reached at the end of that initialization time [5,19]. After the initialization, the ultraviolet light irradiated the suspension and the catalyst was activated. The stirring was stopped and the light was switched off after a period of irradiation. Most particles deposited on the vessel's bottom in a short period. After 10 min, about 1 ml upper solution was aspirated by microsyringe and centrifuged. The concentration of Methylene blue in the supernatant was calculated according to the regression function from the absorption value (ABS) of the working curve

which established the relationship between the absorption and the concentration of Methylene blue in the solution. The ABS for each supernatant after a period of photo-decomposition was obtained from Sefi IUV-1240 UV–vis spectrophotometer (As One Co., Ltd., Japan) at the maximum absorptive wavelength which was 660 nm for Methylene blue [23]. The working curve can be drawn by measuring the absorption value (ABS) of the Methylene blue solution with known concentration.

3. Results and discussion

3.1. Characterization of phase change

Fig. 1 showed the X-ray diffraction patterns of commercial photo-catalytic powder (ST-01) after one hour precalcination at different temperatures. It could be found from Fig. 1, when the temperature was lower than 650 °C the intensity of the peaks on the XRD patterns attributed to anatase steadily became stronger and the width of XRD diffraction peaks attributed to anatase became slightly narrower with the increase of temperature. These characteristics depicted the enhancement of crystallization of anatase with the increase of temperature. When the sample was treated at 850 °C, the phase of the powder changed from anatase to rutile thoroughly.

Fig. 2 showed the X-ray diffraction patterns of the composite granules after the heat treatment at different temperatures for 1 h. At the temperature range from 350 °C to 650 °C, there was no significant change of the diffraction peaks of α-TCP among the granules. But the first most intensive diffraction peak of anatase, which is located at 2θ 25.281° (PDF#21-1272), became distinctively stronger after the heat treatment at the temperature of 850 °C. It could be easily drawn the conclusion that the crystallinity of anatase was enhanced with the increasing of temperature for the composite granules. At the same time, it could be noticed that a new intensive diffraction peak appeared at the 2θ around 27.5° after heat treatment at the temperature of 850 °C. Because the most intensive diffraction peak of rutile is located at 27.446° (PDF#21-1276), and it is almost the third intensive diffraction peak of β-TCP (PDF#09-0169), detail analysis is needed for the attribution of the new appeared diffraction peak. If this peak was attributed to the formation of rutile from anatase, the third intensive diffraction peak of rutile located at 2θ 36.085° with the relative intensity of 50% should be observed at the diffraction pattern. In fact, the diffraction peak of anatase was so intensive, and there were a lot of peaks attributed to β-TCP, it is reasonable to draw the conclusion that the amount of rutile transformed from anatase after the heat treatment at 850 °C of the granule was so small if any that cannot be detected by X-ray diffraction. The formation of β-TCP could also be identified because the intensive of diffraction peaks attributed to α-TCP became weaker after the heat treatment at that temperature. β-TCP is the low temperature morph of $Ca_3(PO_4)_2$ and the transition temperature from β-TCP to α-TCP is about 1125 °C. This phenomenon coincided with others' reports that after lower temperature and longer time heat treatment the α-TCP will transform to β-TCP. Fig. 2 indicated that with the mixture of α-TCP, the phase transformation from anatase to rutile was inhibited and the crystallinity of anatase was enhanced which was preferred to photo-catalytic titanium dioxide and this will be discussed later.

Fig. 3 was the X-ray diffraction pattern of the granule after hydrothermal treatment at 120 °C for 24 h. The X-ray diffraction patterns of the granules precalcined at the temperature range from 350 °C to 850 °C were almost the same. The results indicated that after hydrothermal treatment the α-TCP (some part of β-TCP, especially for those precalcined at 850 °C) was hydrolyzed and the

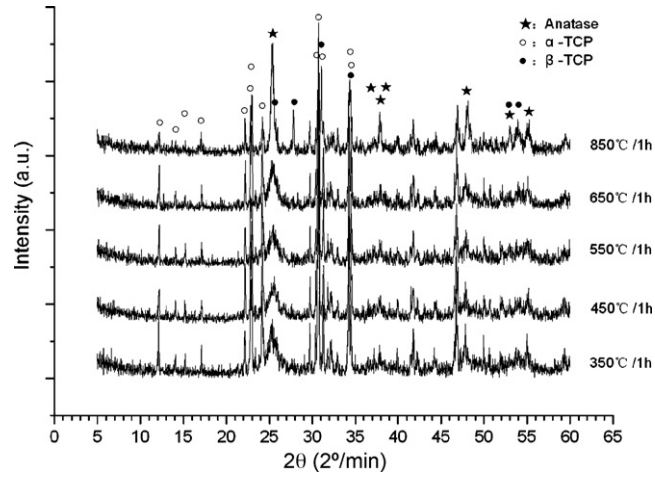

Fig. 2. X-ray diffraction patterns of granule with the composition of titania (30%) and α-TCP (70%) after the heat treatment at different temperatures.

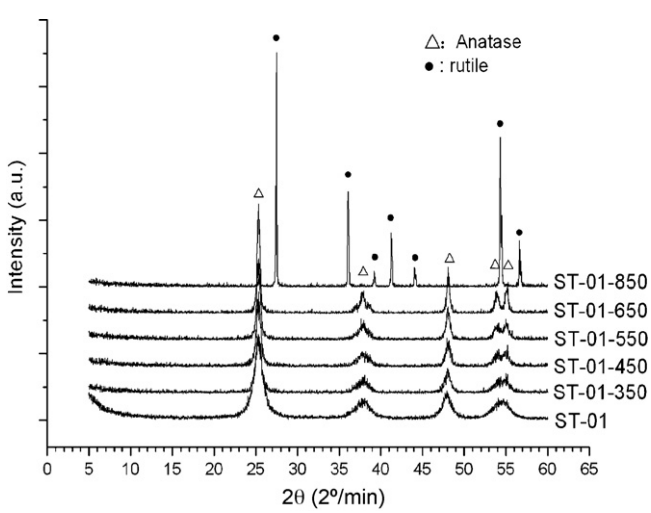

Fig. 1. X-ray diffraction patterns of commercial photo-catalyst powder (ST-01) after one hour precalcination at different temperatures (the precalcination temperature was tagged at the end of 'ST-01' in the legend).

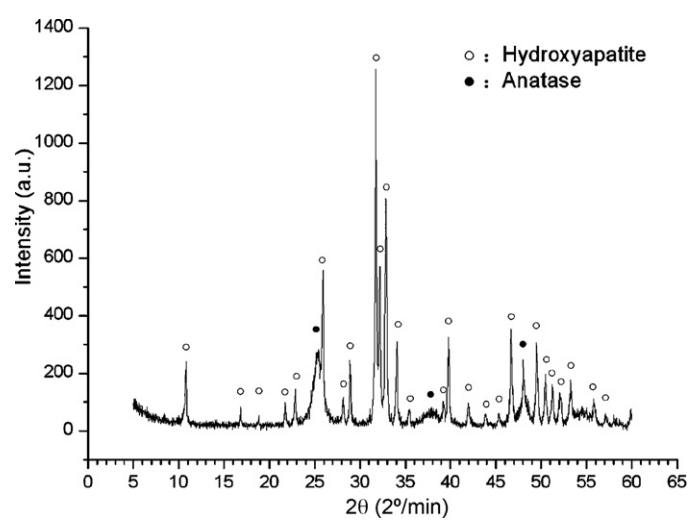

Fig. 3. X-ray diffraction pattern of the composite granule after hydrothermal treatment at 120 °C for 24 h.

HAp was formed. And the results also supported the pervious conclusion that with the mixture of α-TCP, the anatase phase could be remained in higher temperature.

3.2. Analyzing the change of microscopic structure

The optical photograph of the granules prepared under the oil temperature 40 °C with the stirring rate of 300 rpm was shown in Fig. 4(a). The size of the granules distributed from 40 μm to 200 μm in a very round shape. The authors [21] and others [20,22] reported the fabrication of these kinds of granules with different particle size distribution by the alternative of oil temperature and stirring rate. In general, for granules as photo-catalyst, the smaller the size, the stronger the photo-catalytic activity would emerge. But the smaller size is a drawback for recycling as it will take a long time to deposit or cause the lost of catalysts. So it is important to reach the balance between the size distribution of granules and photo-decomposition ability.

Fig. 4(b) showed the SEM image of titania/α-TCP composite granule after the removal of gelatin. It can be found clearly that the granule was highly porous and nano-sized TiO_2 (ST-01) bonded on the surface of coarse grains of α-TCP as skeleton after the heat treatment. The immobilization of TiO_2 on the surface of α-TCP also suppressed the phase transformation of TiO_2 from anatase to rutile.

The SEM image of HAp transformed from α-TCP by hydrolysis and crystallization during the hydrothermal treatment [24] was showed in Fig. 4(c). The results revealed that the spherical shape of granule was retained after hydrothermal treatment, but the skeleton was replaced by rod-shaped HAp for α-TCP particles. The higher magnification SEM image was shown in Fig. 4(d). The aspect ratio of rod-shaped HAp was about 80 with the average length of 24 μm and the average sectional size of 0.3 μm. It can also be noticed that most rod-shaped HAp particles clustered to form coarse aggregates.

Fig. 4(e) was the image of the microstructure of titania/α-TCP composite granule after hydrothermal treatment at 120 °C for 24 h. Compared with Fig. 4(d), in the composite granule the length of rod-shaped HAp became shorter distinctively but the diameter almost kept unchanged. The aspect ratio decreased sharply from 80 to 10. A large amount of previous literatures reported the formation of HAp elongated along the *c*-axis of HAp from α-TCP powder by hydrothermal treatment. Since the crystallization transition was hard to accomplish at solid state, during hydrothermal treatment process in relative lower temperature we proposed that a process involving dissolution–crystallization might account for this transformation. α-TCP has high reactivity in water with the thermodynamic solubility product k_{sp} of 0.22×10^{-29} [25] and it will hydrolyze to HAp whose thermodynamic solubility product k_{sp} is 2.12×10^{-59} [26] following the mechanism mentioned above. When α-TCP was mixed with titania, the *c*-axis elongated growth of HAp during hydrothermal treatment process was halted and the aggregation of rod-shaped

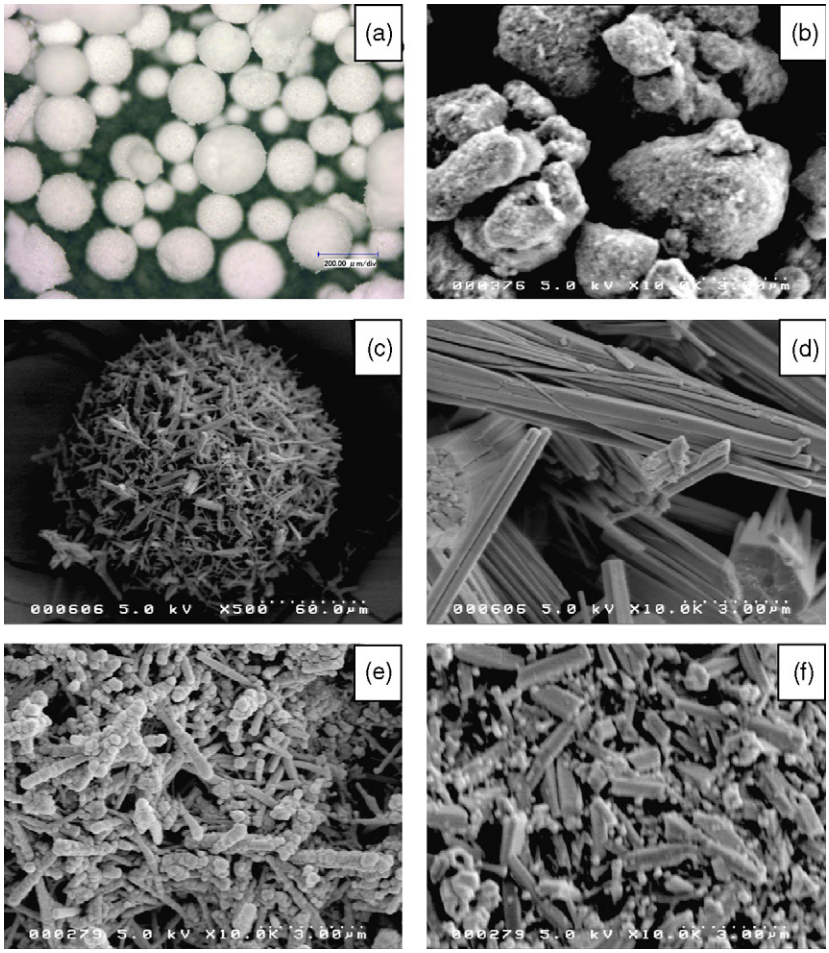

Fig. 4. (a) Optical photograph of the granules (oil temperature: 40 °C, stirring rate: 300 rpm); (b) SEM image of titania/α-TCP composite granule after the removal of gelatin by precalculation; (c) SEM image of α-TCP granule after hydrothermal treatment at 120 °C for 24 h; (d) high magnification SEM image of (c); (e) SEM image of titania/α-TCP composite granule after hydrothermal treatment at 120 °C for 24 h; (f) SEM image of compressed titania/α-TCP composite powder after hydrothermal treatment at 120 °C for 24 h.

Table 1
S_{BET} and specific surface area retaining rate of anatase for different samples.

Item	ST-01	ST-01-550	HAp granule	α-TCP granule	Titania/HAp granule	Compressed powder
S_{BET} (m^2/g)	335.98	87.68	0.71	0.11	35.25	17.54
Retaining rate[a] (%)	100	26.09	–	–	34.97[b]	17.40[c]

[a] Retaining rate = (S_{BET} of ST-01 after treatment)/(S_{BET} of commercial ST-01).
[b] Titania/HAp composite granule containing 30 wt.% ST-01, removing gelatin at 550 °C for 1 h.
[c] Compressed powder containing 30 wt.% ST-01.

HAp became less. At the same time the nano-sized anatase was anchored on the surface of rod-shaped HAp effectively.

Fig. 4(f) showed the SEM micrograph of compressed titania/α-TCP composite powder after hydrothermal treatment. Even though the constituent was the same with titania/α-TCP composite granule, the short-pillar-shaped HAp was formed after hydrothermal treatment. Beside this, the anchoring of anatase on the surface of HAp was seldom found in this kind of structure. Most anatase aggregated in the interval of short-pillar-shaped HAp. These may be explained by the difference of fabrication processes adopted for compressed powder and granule. Because the mixed powder was pressed under the pressure of 200 MPa, after crushing, the compressed powder showed less porosity compared with granule formed in oil. The kinetic conditions were different from granule to compressed powder during the hydrothermal treatment process.

Table 1 listed the S_{BET} value and specific surface area retaining rate for these samples. The result could be well explained by their microstructure. α-TCP and HAp granules had very low specific surface area. The results of S_{BET} value for ST-01-550, titania/HAp granule and compressed powder indicated that the composite granule would provide active anatase surface for photo-catalytic reaction in great extent.

The diffuse reflectance UV–vis spectra of the titania/HAp, ST-01 and HAp were shown in Fig. 5. As expected, ST-01 (mainly anatase) and HAp has no absorption above their fundamental absorption sharp edge. The band gaps of anatase and HAp were 3.2 eV and 5.3 eV, respectively. The differential curves ($dR/d\lambda \sim \lambda$) were consistent with these very well. The peak appeared at 386 nm for ST-01 and there was no absorption above 250 nm for HAp. On the contrary, the HAp showed its high reflective property at the wave range between 250 nm and 350 nm. A 16-nm blue-shift of the absorption edge for the titania/HAp composite granule could be noticed in our experiment.

3.3. Evaluation of photo-catalytic activity

Fig. 6 showed the results of photo-catalytic decomposition of Methylene blue by composite granules, compressed powder and ST-01 (with and without precalcination). The amount of anatase was the same for each sample as described before. It has been proved in our experiment that the adsorption of Methylene blue by ST-01 was so weak and there was almost no obvious concentration decrease of Methylene blue caused by the adsorption of ST-01 after long time treatment. And no further decrease of the concentration of Methylene blue could be checked after the initial time for HAp and titania/HAp composite in dark. It meant that the adsorptive saturation was reached after the initial time. It was evident from Fig. 6, the concentration decrease of Methylene blue caused by adsorption was less than 0.5 ppm for all granular samples and there was no photo-decomposition in the dark. Only after the irradiation started, the decomposition of Methylene blue initiated. The results indicated that the composite granules showed very high decomposition ability. More than 85% of Methylene blue was decomposed after one hour irradiation and it was almost thoroughly decomposed after two hour irradiation.

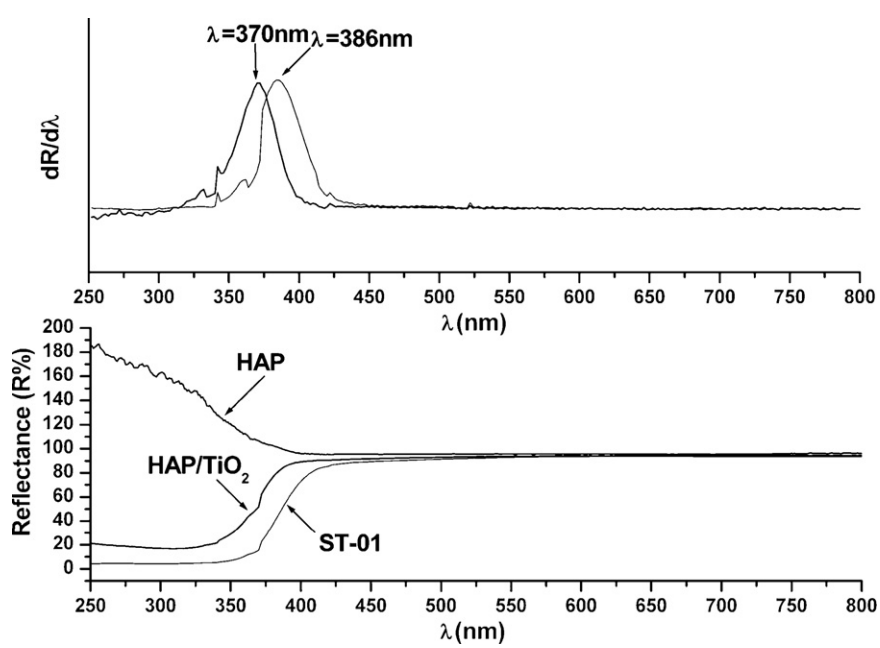

Fig. 5. Diffuse reflectance UV–vis spectra of titania/HAP composite granule, ST-01 and HAP powder.

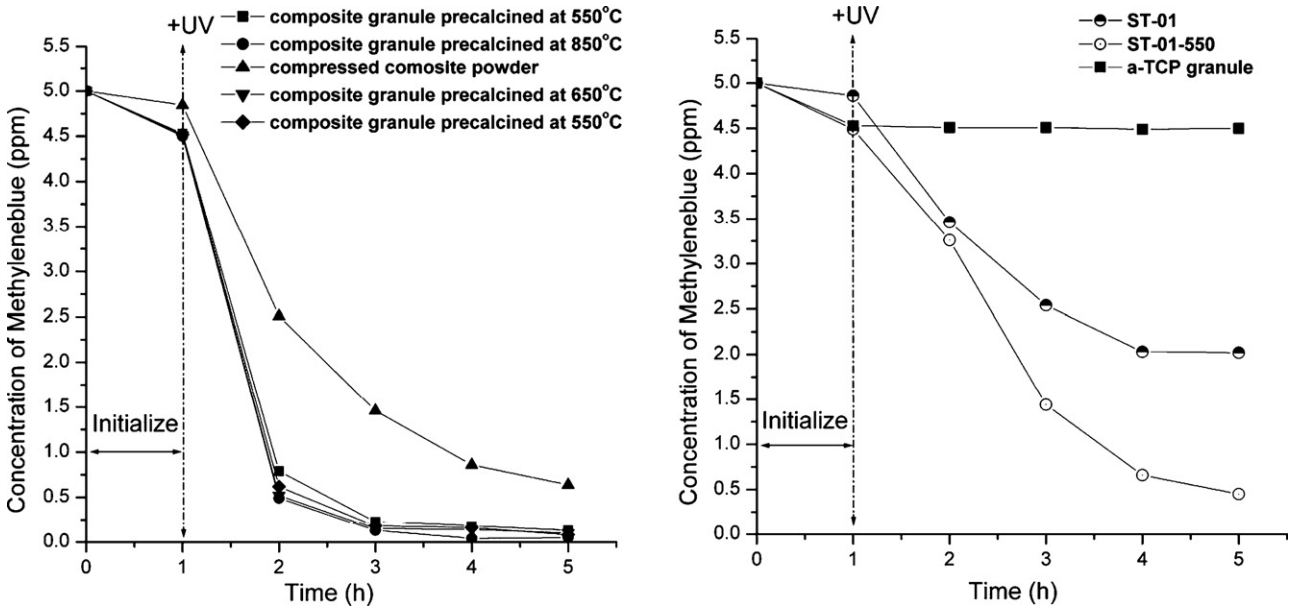

Fig. 6. Diagram of concentration of MB against time of photo-catalytic decomposition of Methylene blue by granule, compressed powder and ST-01.

For quantitative analysis of the decomposition of Methylene blue, the first-order kinetic model was employed

$$-\frac{dc}{dt} = kc$$

where k is the observed rate constant for the photo-decomposition reaction.

From the above equation, the following equation can be obtained:

$$\ln c = -kt + \ln C_0$$

The concentration of Methylene blue in the solution ($\ln C$) was plotted against irradiation time in Fig. 7. The good linear agreement between $\ln c$ and t for granule, compressed powder, ST-01 and ST-01-550 suggested that the present reaction follows the first-order kinetic. Table 2 listed the reaction rate constants calculated from the slope of these lines. It was very clear that the granular type photo-catalyst sample owned the highest photo-activity for the decomposition of Methylene blue. The observed rate constant of granular type photo-catalyst was more than five times compared with commercial photo-catalyst ST-01.

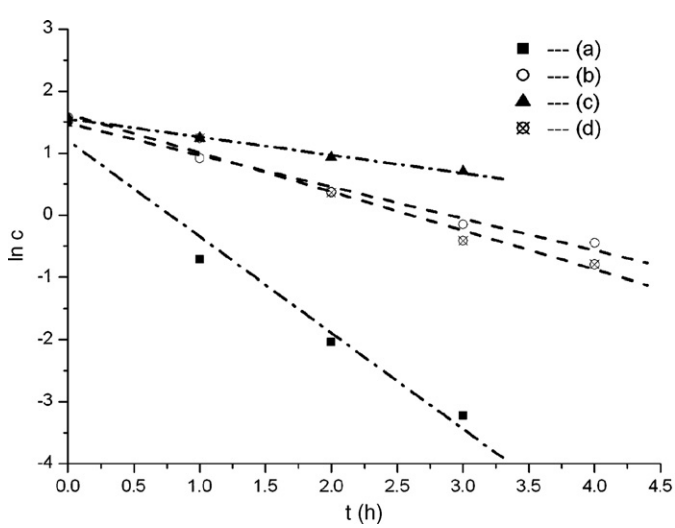

Fig. 7. Plots of ln C against irradiation time of ultraviolet for four kinds of samples ((a) composite granule, (b) compressed powder, (c) ST-01, and (d) ST-01-550).

It could also be found that there was slight difference of decomposition ability for granules precalcined in different temperatures. With the increase of precalcination temperature the photo-catalytic activity was enhanced, but in a small extent for these composites. Since the difference of the resulted micro-structure by hydrothermal treatment of titania/α-TCP composite samples precalicined at different temperatures was not so distinct, this slight difference of decomposition ability for granules might be attributed to the enhancement of crystallinity of anatase with the increase of precalcination temperature.

This point of view could be strongly supported by comparing the decomposition ability of ST-01 and ST-01-550 (precalcined ST-01 at 550 °C for 1 h). Even though the S_{BET} value of ST-01-550 was only a quarter of ST-01, the increase of crystallinity of anatase compensated the decrease of specific surface area and enhanced the photo-catalytic ability as shown in Fig. 6. After precalcination the observed rate constant for the photo-decomposition reaction of ST-01 increased by 117%. Titanium dioxide has three allotropies, i.e. anatase, rutile and brookite. However only anatase and rutile phases have been used in most photo-catalytic investigations and the photo-catalytic activity of amorphous titania or brookite is negligible [27]. Anatase phase shows higher photo-catalytic activity than rutile due to its low recombination rate of photo-generated hole–electron pairs formed on the cause of the leap of electron from valence band to the conduction band and its wider band gap. So the high crystallinity and perfect of the crystal were favorable for anatase as photo-catalytic material.

Table 2
Observed reaction rate constant (k) of different samples for photo-decomposition of Methylene blue.

Photo-catalyst sample	k (ppm/h)
Composite granule	1.55
Compressed powder	0.51
ST-01	0.29
ST-01-550	0.63

The hybridization of ST-01 and HAp could also bring about two other effects. One was the blue-shift of the absorption edge; another was the enhancement of the separation of the photo-induced charges. These should partly enhance the decomposition ability of titania/HAp composite to Methylene blue. The blue-shift of the absorption edge of the composite deduced the photo-generated hole with higher oxidative potential and the photo-generated electron with higher reductive capability. It was reported that *a*-face of the *c*-axis elongated HAp was positive charged and the side surface of the HAp crystal was negative charged. The positive charge should be balanced by anions in the solution. In our experiment, the OH^- was the most possible one to balance the positive surface charge. From the previous analysis, the positive charged HAp surface was covered by nano-sized ST-01. The concentration of OH^- was higher in the surround of ST-01 than other areas in the solution. When the photon was absorbed by TiO_2, the electron and hole could be excited in the following reaction:

$$TiO_2 + h\nu \rightarrow TiO_2 + h^+ + e^-$$

The photo-induced electron and hole could recombine. In other way, this recombination could be retarded or halted by the following reaction:

$$h^+ + OH^- \rightarrow {}^{\bullet}OH$$

The hole-produced hydroxyl radicals (${}^{\bullet}OH$) play the most important role in most liquid–solid photo-catalysis oxidative process.

It is well known that the photo-catalytic reaction happened around the surface area. So the surface area is another important factor influencing catalytic reaction. In Table 2, the observed rate constant for the photo-decomposition reaction of composite granule increased by 142% compared with ST-01-550. This great obviously boost could be ascribed to the effectively retaining of the specific surface area of anatase in composite granule. The granule owning the microstructure of Fig. 4(e) provided a large amount of active surface of anatase and worked as three-dimensional highly porous small reactor by anchoring nano-sized anatase on the surface of rod-shaped dispersed HAp. From Table 1, it could be concluded that the S_{BET} value of composite granule should be mostly ascribed to fine anatase particles. The effective active surface area of anatase in composite granule was greater than that of ST-01-550. On the other hand, when fine anatase powder was used directly to decompose Methylene blue in solution, the effective surface area would decrease for the aggregation of nano-sized fine particles. But by the immobilization of nano-sized anatase particles on the surface of rod-shaped HAp, this phenomenon could be effectively eliminated.

4. Conclusions

The size-controllable granules composed of titania and α-TCP were fabricated by the process based on liquid immiscibility effect. The precalcination for the removal of gelatin increased the crystallinity of anatase and partly enhanced the photo-catalytic decomposition ability to Methylene blue. It could be confirmed from X-ray diffraction patterns that the mixture of α-TCP in the anatase powder suppressed the phase transformation from anatase to rutile effectively.

High porosity of the fabricated granule provided good condition for the formation of an ideal microstructure which can be described as that the granule was constructed by the skeleton of rod-shaped HAp and the surface of HAp was anchored by nano-sized anatase particles during the dissolution–crystallization process by environmentally friendly hydrothermal treatment. After hydrothermal treatment process the active surface of anatase was effectively retained and the aggregation phenomena that usually happened when use fine anatase directly in solution was eliminated. The hybridization of titania and HAp leads the blue-shift of the absorption edge and the enhancement of the separation of the photo-induced charges. Quantitative analysis of the decomposition reaction indicated that the observed photo-catalytic reaction rate constant of the fabricated granule by hydrothermal treatment was greatly increased.

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